

Ethan Reid

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EDUCATION

B.S. Economics | Quantitative Analysis Focus | Computer Science Minor San Luis Obispo, CA | June 2023

CALIFORNIA POLYTECHNIC STATE UNIVERSITY

CS GPA: 3.94/4.0

Coursework: Numerical Analysis (UC Berkeley), Machine Learning for Economics (Graduate), Systems Programming, Advanced Econometrics, Probability Models, **Linear Optimization (Economics)**, Mathematical Economics (Differential Equations), Statistical Computing, Discrete Structures

WORK EXPERIENCE

MOONDREAM | FOUNDING RESEARCH SCIENTIST

San Francisco, CA | October 2024 – Present

- Co-led architecture development of Moondream Segmentation, Moondream 3, 2, and 0.5B.
- Invented a rank-pruning algorithm that produced Moondream 0.5B.
- Designed pretraining, SFT, and RL infrastructure.

CHEVRON | QUANTITATIVE SOFTWARE ENGINEER 2

San Ramon, CA / Houston, TX | July 2023 – October 2024

- Developed a parallel simulated-annealing optimizer in Python for Supply and Trading, generating **25MM/year** by identifying profitable short positions, refining routing, and assessing regional auctions.
- Built a high-performance LP optimizer in Numba, **improving runtime by 15x** and enabling its use in short-term trading.

BLUCHIP | MACHINE LEARNING INTERN

San Diego, CA | June 2022 – September 2022

- Designed **XGBoost**-based model to forecast success of potential new commercial properties. Developed a system in Python to monitor relevance of the model, augmenting and retrieving new data to prevent obsolescence. Deployed on AWS.
- Developed a load management system to route inference/training tasks between instances to ensure performant use of GPUs.

NOMAD | CO-FOUNDER & CEO

San Francisco, CA | 2016 – 2018

- With a team of 12, constructed a YOLO-based computer vision platform to enable continuous training and inference via a web GUI and API.
- Supported by Hewlett Packard Enterprise and a **focus in the 2017 documentary, Seed**.

MODELS

MOONDREAM SEGMENTATION 

2025

Invented **state-of-the-art Referring Segmentation model**, outperforming SAM3, Gemini 2.5 Pro, and Kimi SimpleSeg. Novel two-stage process where the model first autoregressively generates a vector path, then refines the coarse rasterization. Technical report under review for ICML 2026.

MOONDREAM 3 

2025

2A9B Mixture of Experts VLM achieving new highs on vision benchmarks. Exceeds GPT5 and Claude 4 on vision grounding tasks. CuTe DSL kernels enable 30fps on H100.

MOONDREAM 2 

2024

One of the first and most popular VLMs. **Over 30MM downloads on HF**.

INTERESTS

Research: VLMs, Optimizers, Diffusion, World Models

Frameworks: PyTorch, JAX, FSDP, CuTe DSL, CUDA

Hobbies: Hackathons (15+ Wins), Skiing, Mountain Biking/Cycling (College Team), Golf